

Ninghao Liu

CONTACT INFORMATION	Computer Science and Engineering 518 H.R. Bright Building College Station, TX 77843, USA	(404) 719-1028 nhliu43@tamu.edu http://people.tamu.edu/~nhliu43
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OBJECTIVE

To look for a tenure-track assistant professor position.

RESEARCH INTERESTS

- **Explainable Artificial Intelligence (XAI):** Post-Hoc Model Interpretation, Representation Interpretation, Interpretable Modeling, Health Informatics
- **Data Mining:** Network Representation Learning, Recommender Systems, Graph Convolutional Networks
- **Information Security:** Anomaly Detection, Adversarial Machine Learning, Backdoor Attacks

EDUCATIONAL BACKGROUND

- Texas A&M University Ph.D., Computer Science Jan 2016 - May 2021 (expected)
 - *Dissertation: Extracting and Harnessing Interpretation in Data Mining*
 - *Advisor: Dr. Xia “Ben” Hu*
 - *Committee members: Dr. Xia Hu, Dr. James Caverlee, Dr. Nick Duffield, Dr. Ruihong Huang*
- Georgia Institute of Technology M.S., Electrical and Computer Engineering Aug 2014 - Dec 2015
- South China University of Technology BEng., Information Engineering (Pilot Class) Sept 2010 - July 2014
 - *GPA: Top 1% among 400+ students*

PUBLICATIONS

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| Summary | 33 publications: 19 conference papers, 2 book chapter, 11 journal papers, & 1 arXiv preprints
12 first-author papers and 20+ collaborators
Published in various top-notch conferences: KDD, WWW, WSDM, CIKM, IJCAI, ICDM, SDM
550+ citations, h-index 11 , i10-index 14 |
| Awards | [C6]: Best Paper Award Candidate, ICDM 2019
[C8]: Best Paper Award Shortlist, WWW 2019
[C16]: Best Paper Award, IEEE ICCS 2014 |
| Research Areas and Representatives | <ul style="list-style-type: none">• Post-Hoc Model Interpretation: [C6], [C7], [C8], [C10], [C12], [C13], [C14], [J1], [J5]<ul style="list-style-type: none">– Ninghao Liu, Xiao Huang, Jundong Li, Xia Hu. “On Interpretation of Network Embedding via Taxonomy Induction”, <i>ACM SIGKDD International Conference on Knowledge Discovery and Data Mining</i>, 2019.• Network Data Mining: [C1], [C5], [C10], [J2]<ul style="list-style-type: none">– Ninghao Liu, Qiaoyu Tan, Yuening Li, Hongxia Yang, Jingren Zhou, Xia Hu. “Is a Single Vector Enough? Exploring Node Polysemy for Network Embedding”, <i>ACM SIGKDD International Conference on Knowledge Discovery and Data Mining</i>, 2019.• Information Security: [C2], [C6], [C9], [C11], [C14], [C15], [J1], [A1]<ul style="list-style-type: none">– Ninghao Liu, Hongxia Yang, Xia Hu. “Adversarial Detection with Model Interpretation”, <i>ACM SIGKDD International Conference on Knowledge Discovery and Data Mining</i>, 2018.• Recommender Systems: [C1], [C3], [C5], [C13]<ul style="list-style-type: none">– Ninghao Liu, Yong Ge, Li Li, Xia Hu, Rui Chen, Soo-Hyun Choi. “Explainable Recommender Systems via Resolving Learning Representations”, <i>The Conference on Information and Knowledge Management</i>, 2020. |

Refereed Conference Publications

- [C1] Qiaoyu Tan, Jianwei Zhang, **Ninghao Liu**, Xiao Huang, Hongxia Yang, Jingren Zhou, Xia Hu. “Dynamic Memory based Attention Network for Sequential Recommendation”, *AAAI Conference on Artificial Intelligence (AAAI)*, 2021
- [C2] Qiaoyu Tan, Jianwei Zhang, Jiangchao Yao, **Ninghao Liu**, Jingren Zhou, Hongxia Yang, Xia Hu. “Sparse-Interest Network for Sequential Recommendation”, *ACM International Conference on Web Search and Data Mining (WSDM)*, 2021
- [C3] **Ninghao Liu**, Yong Ge, Li Li, Xia Hu, Rui Chen, Soo-Hyun Choi. “Explainable Recommender Systems via Resolving Learning Representations”, *The Conference on Information and Knowledge Management (CIKM)*, 2020
- [C4] Kion Fallah, Adam Willats, **Ninghao Liu**, Christopher Rozell. “Learning Sparse Codes From Compressed Representations With Biologically Plausible Local Wiring Constraints”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2020
- [C5] Qiaoyu Tan, **Ninghao Liu**, Hongxia Yang, Jingren Zhou, Xia Hu. “Learning to Hash with Graph Neural Networks for Recommender Systems”, *The Web Conference (WWW)*, 2020
- [C6] Ruixiang Tang, Mengnan Du, **Ninghao Liu**, Fan Yang and Xia Hu. “An Embarrassingly Simple Approach for Trojan Attack in Deep Neural Networks”, *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020
- [C7] Fan Yang, **Ninghao Liu**, Mengnan Du, Kaixiong Zhou, Shuiwang Ji, Xia Hu. “Deep Neural Networks with Knowledge Instillation”, *SIAM International Conference on Data Mining (SDM)*, 2020
- [C8] **Ninghao Liu**, Qiaoyu Tan, Yuening Li, Hongxia Yang, Jingren Zhou, Xia Hu. “Is a Single Vector Enough? Exploring Node Polysemy for Network Embedding”, *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2019
- [C9] Mengnan Du, **Ninghao Liu**, Fan Yang, Xia Hu. “Learning Credible Deep Neural Networks with Rationale Regularization”, *IEEE International Conference on Data Mining (ICDM)*, 2019
(**Best Paper Award Candidate, ICDM 2019**)
- [C10] **Ninghao Liu**, Mengnan Du and Xia Hu. “Representation Interpretation with Spatial Encoding and Multimodal Analytics”, *ACM International Conference on Web Search and Data Mining (WSDM)*, 2019
- [C11] Mengnan Du, **Ninghao Liu**, Fan Yang, Shuiwang Ji, Xia Hu. “On Attribution of Recurrent Neural Network Predictions via Additive Decomposition”, *The Web Conference (WWW)*, 2019
(**Best Paper Award Shortlist, WWW 2019**)
- [C12] Yuening Li, **Ninghao Liu**, Jundong Li, Mengnan Du, Xia Hu. “Deep Structured Cross-Modal Anomaly Detection”, *International Joint Conference on Neural Networks (IJCNN)*, 2019
- [C13] **Ninghao Liu**, Xiao Huang, Jundong Li, Xia Hu. “On Interpretation of Network Embedding via Taxonomy Induction”, *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2018
- [C14] **Ninghao Liu**, Hongxia Yang, Xia Hu. “Adversarial Detection with Model Interpretation”, *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2018
- [C15] Mengnan Du, **Ninghao Liu**, Qingquan Song, Xia Hu. “Towards Explanation of DNN-based Prediction with Guided Feature Inversion”, *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2018
- [C16] Fan Yang, **Ninghao Liu**, Suhang Wang, Xia Hu. “Towards Interpretation of Recommender Systems with Sorted Explanation Paths”, *IEEE International Conference on Data Mining (ICDM)*, 2018
- [C17] **Ninghao Liu**, Donghwa Shin, Xia Hu. “Contextual Outlier Interpretation”, *International Joint Conference on Artificial Intelligence (IJCAI)*, 2018

- [C18] **Ninghao Liu**, Xiao Huang, Xia Hu. “Accelerated Local Anomaly Detection in Attributed Networks”, *International Joint Conference on Artificial Intelligence (IJCAI)*, 2017
- [C19] Jing Lu, Zuming Huang, **Ninghao Liu**, and Quansheng Guan. “Distance-weighted backlog differentials for back-pressure routing in multi-hop wireless networks”, *IEEE/CIC International Conference on Communications in China*, 2014.
(Best Paper Award, IEEE ICC 2014)

Book Chapters

- [A1] **Ninghao Liu**, Qizhang Feng, Xia Hu, “Interpretability in Graph Neural Networks”. In Lingfei Wu, Peng Cui, Liang Zhao and Jian Pei. *Graph Neural Networks: Foundations, Frontiers, and Applications*. forthcoming with Springer, 2021.
- [A2] **Ninghao Liu**, Xia Hu, “Spam detection in online social media: A survey”. In Jiliang Tang and Charu Aggaral. *Social Media Analytics: Advances and Applications*. forthcoming with CRC Press, 2020.

Refereed Journal Publications

- [J1] **Ninghao Liu**, Mengnan Du, Ruocheng Guo, Huan Liu, Xia Hu, “Adversarial Attacks and Defenses: An Interpretation Perspective”, SIGKDD Exploration Newsletter, 2021 (to appear in the June issue).
- [J2] Weijie Fu, Meng Wang, Mengnan Du, **Ninghao Liu**, Shijie Hao, Xia Hu, “Differentiated Explanation of Deep Neural Networks with Skewed Distributions”, *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2020.
- [J3] Mengnan Du, **Ninghao Liu**, Fan Yang, Xia Hu. “Learning Credible DNNs via Incorporating Prior Knowledge and Model Local Explanation”, *Knowledge and Information Systems (KAIS)*, 2020
- [J4] Mengnan Du, **Ninghao Liu**, Xia Hu. “Techniques for Interpretable Machine Learning”, *Communications of the ACM*, 2019
(The Highlighted Article on the cover page of the January 2020 issue, *Communications of the ACM*)
- [J5] Qiaoyu Tan, **Ninghao Liu**, Xia Hu. “Deep Representation Learning for Social Network Analysis”, *Frontiers in Big Data*, 2019
- [J6] Nur Hafeza Ismail, **Ninghao Liu**, Mengnan Du, Zhe He, Xia Hu, “Identification of Cancer Survivors Living with PTSD on Social Media”, *Studies in health technology and informatics*, 2019
- [J7] **Ninghao Liu**, Xia Hu. “Spam Detection on Social Networks”, *Encyclopedia of Social Network Analysis and Mining*, 2017
- [J8] Jun Gao, **Ninghao Liu**, Mark Lawley, Xia Hu. “An Interpretable Classification Framework for Information Extraction from Online Healthcare Forums”, *Journal of Healthcare Engineering*, 2017
- [J9] Xia Hu, Peter Reaven, Aramesh Saremi, **Ninghao Liu**, Mohammed Ali Abbasi, Huan Liu, Raymond Migrino. “Machine learning to predict rapid progression of carotid atherosclerosis in patients with impaired glucose tolerance”, *EURASIP Journal on Bioinformatics and Systems Biology*, 2016
- [J10] Xiangyu Yu, **Ninghao Liu**, Xin Qian, Tao Zhang. “Deployment Method Based on Spring Force in Wireless Robot Sensor Networks”, *International Journal of Advanced Robotic Systems*, 2014
- [J11] Xiangyu Yu, **Ninghao Liu**, Weipeng Huang, Xin Qian, Tao Zhang. “A Node Deployment Algorithm Based on Van Der Waals Force in Wireless Sensor Networks”, *International Journal of Distributed Sensor Networks*, 2013

ArXiv Preprint

- [A1] **Ninghao Liu**, Yunsong Meng, Xia Hu, Tie Wang, Bo Long, “Are Interpretations Fairly Evaluated? A Definition Driven Pipeline for Post-Hoc Interpretability”, arXiv preprint arXiv:2009.07494, 2020.

RESEARCH EXPERIENCE

- Research internship at Samsung Research America June 2019 - Aug 2019
 - *Developed a novel recommendation model based on graph neural networks to capture diverse user interests.*
 - *Published a research paper [C1] in CIKM 2020.*
- Visiting student at Alibaba DAMO Academy June 2018 - Aug 2018
 - *Developed multi-aspect network embedding by considering node polysemy for network embedding.*
 - *Published a research paper [C6] in KDD 2019*
- Research assistant, Texas A&M University, College Station Jan 2016 - present
 - *Explored how to open the black-boxes of machine learning and deep learning models with applications in information security and health informatics.*
 - *Member of a NSF Project: Adversarial Learning via Modeling Interpretation.*
 - *Member of a DARPA Project: Transforming Deep Learning to Harness the Interpretability of Shallow Models: An Interactive End-to-End System.*
- The Fundamental Research Project for Central Universities May 2012 - Dec 2013
 - *Designed a system for robust face recognition using OpenCV.*

PROPOSAL WRITING

- III: Medium: Collaborative Research: Towards Effective Interpretation of Deep Learning: Prediction, Representation, Modeling and Utilization (PI: Dr. Xia Hu)
 - *Role: Led efforts to organize and coordinate the writing of the four research objectives, and significantly contributed to two of them, including “Interpretation of Deep Representation” and “Design of Interpretable Deep Models”.*
 - *Result: Submitted to National Science Foundation (NSF) and got funded in August 2019.*
- SaTC: CORE: Small: Adversarial Learning via Modeling Interpretation (PI: Dr. Xia Hu)
 - *Role: Led efforts to organize and coordinate the writing of the three research objectives, and significantly contributed to two of them, including “Interpretation-based Attacks” and “Defensive Strategies”.*
 - *Result: Submitted to National Science Foundation (NSF) and got funded in July 2018*
- CAREER: Human-Centric Big Network Embedding (PI: Dr. Xia Hu)
 - *Role: Contributed to one of the three research objectives, “Interpretable and Interactive Network Embedding”*
 - *Result: Submitted to National Science Foundation (NSF) and got funded in July 2018*
- Transforming Deep Learning to Harness the Interpretability of Shallow Models: An Interactive End-to-End System (PI: Dr. Xia Hu)
 - *Role: Contributed to one of the three research objectives, “Mining Patterns based on Shallow Models”*
 - *Result: Submitted to DOD-Advanced Research Projects Agency (DARPA) and got funded in 2017.*

TEACHING AND TALKS

- Teaching Assistant at Texas A&M University Jan 2017 - May 2017
 - *Undergraduate course: Information Storage and Retrieval*

MENTORING

- Hongye Jin, Ph.D. in Computer Science, TAMU Sept 2020 - Present
 - *Co-authored 1 paper submitted to KDD*
- Yushun Dong, Ph.D. in Computer Science, University of Virginia Sept 2020 - Present
 - *Working on a project of GNN fairness*
- Ruixiang Tang, Ph.D. in Computer Science, TAMU Aug 2019 - Jan 2021

- Co-authored 1 paper accepted by KDD2020, 1 paper submitted to KDD2021
- Qiaoyu Tan, Ph.D. in Computer Science, TAMU Aug 2018 - Dec 2020
 - Co-authored 3 papers at top venues including WWW, WSDM, AAAI
- Nur Hafieza Ismail, Ph.D. in Computer Science, TAMU May 2018 - April 2019
 - Project on "Identification of Cancer Survivors Living with PTSD"
- Fan Yang, Ph.D. in Computer Science, TAMU July 2018 - May 2020
 - Co-authored 2 papers at ICDM and SDM
- Mengnan Du, Ph.D. in Computer Science, TAMU Dec 2017 - May 2020
 - Co-authored 4 papers at top venues including KDD, WWW, ICDM, and CACM
- Jun Gao, M.S. in Computer Science, TAMU May 2017 - Oct 2017
 - Co-authored 1 paper accepted by Journal of Healthcare Engineering

AWARDS & ACTIVITIES

- Conference travel grant to attend CIKM 2020 Sep 2020
- Highlighted Article on the cover page of the January 2020 issue, Communications of the ACM Jan 2020
- 2019 ICDM Best Paper Award Candidate, Beijing, China Nov 2019
- Conference travel grant to attend KDD 2019, Anchorage, USA Aug 2019
- 2019 WWW Best Paper Award Shortlist, San Francisco, USA Apr 2019
- Conference travel grant to attend KDD 2018, London, UK Aug 2018
- "Meritorious Winner" in the Interdisciplinary Contest of Modeling Apr 2013
- National Scholarship Oct 2013
- "Hongping Changqing" Science and Technology Innovation Award Apr 2013
- First Class Scholarship in South China University of Technology Oct 2012
- First Class Scholarship in South China University of Technology Oct 2011

SERVICE

- Program committee member:
 - The 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Singapore, 2021
 - The 38th International Conference on Machine Learning (ICML), Vienna, Austria, 2021
 - The 4th IJCAI-PRICAI 2020 Workshop on XAI, Yohoama, Japan, 2020
 - The 1st Workshop on Artificial Intelligence for Anomalies and Novelities, Yohoama, Japan, 2020
 - The 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, San Diego, CA, 2020
 - The 3rd IJCAI Workshop on Explainable Artificial Intelligence (XAI 19), Macau, China, 2019
 - The 25th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Anchorage, Alaska, 2019
 - The 28th ACM International Conference on Information and Knowledge Management, Beijing, China, 2019 (Long Papers, Short Papers)
 - The 2nd IJCAI Workshop on Explainable Artificial Intelligence (XAI 18), Stockholm, Sweden, 2018
 - The 1st IJCAI Workshop on Explainable Artificial Intelligence (XAI 17), Melbourne, Australia, 2017
- Reviewer:
 - IEEE Transactions on Neural Networks and Learning Systems (TNNLS) (2021)
 - The ACM Transactions on Knowledge Discovery from Data (TKDD) (2020)
 - AMIA 2020 Annual Symposium (2020)
 - Journal of Healthcare Informatics Research (JHIR) (2020)
 - The ACM Transactions on Knowledge Discovery from Data (TKDD) (2019)
 - IEEE Transactions on Neural Networks and Learning Systems (TNNLS) (2019)
 - The ACM Transactions on Knowledge Discovery from Data (TKDD) (2018)

- *Data Mining and Knowledge Discovery (DAMI) (2018)*
- *Applied Stochastic Models in Business and Industry (ASMB) (2018)*
- *Data Mining and Knowledge Discovery (DAMI) (2017)*
- External reviewer:
 - *IEEE International Conference on Data Mining (ICDM 2019)*
 - *Association for the Advancement of Artificial Intelligence (AAAI 2016)*
 - *Conference on Information and Knowledge Management (CIKM 2016)*
 - *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2016)*
 - *International Conference on Social Computing Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS 2016)*